Foundations:
Discrete Maps and Chaos

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guides

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This is a collection of notes based on the discrete-map topics from my Mathematical Methods for Physicists II coursework. It introduces the mathematics needed to understand and implement notebook 4A, with references to earlier notes where appropriate. The companion 4A guide walks through the exercises. Notebook 4B is a separate radioactive-counting extension, whose additional physics and inference concepts are introduced in its own guide.

1 Reading route: maps are deterministic update rules

Notebook 4A studies rules of the form $x_{n+1} = f(x_n; \mu)$, where n is an integer iteration index, x_n is the current state, and μ is a parameter. Once the initial state and parameter are fixed, the rule determines the entire sequence. No random generator is needed.

The 1A/1B probability and random-walk notes and the shared Monte Carlo foundations for 3A/3B describe stochastic sampling. The 2A/2B foundations describe continuous dynamics, numerical integration, and chaos. Here we develop the additional ideas needed for *one-dimensional discrete maps*: invariant intervals, cobwebs, fixed-point multipliers, periodic orbits, and bifurcations.

A map can be a model in its own right, such as population changes between generations. It is not necessarily an approximation to an ODE, and increasing the number of iterations is not the same as decreasing a numerical step size. Notebook 4B is a separate application about stochastic decay counts and parameter inference; its additional concepts belong in its own guide.

2 State, parameter, orbit, and iteration

The state x_n changes at each update. The parameter μ specifies the rule and remains fixed within one trajectory. The initial state x_0 selects the starting point. The sequence $x_0, x_1, x_2, \ldots$ is an orbit in the language of dynamical systems; it need not refer to a spatial planetary orbit.

An implementation should store x_0, repeatedly evaluate the rule at the latest stored state, and append each new value. Taking N updates produces $N+1$ states. In Python, a loop starting at zero must still distinguish its loop counter from the number of completed updates.

Iterating means composing a function with itself: $x_2 = f(f(x_0))$. It does not mean squaring its value, multiplying by the iteration index, or reevaluating at x_0 every time. In notation, $f^{\circ p}$ will mean p compositions, avoiding confusion with an ordinary power.

3 The logistic population model

The logistic map is

$$x_{n+1} = \mu x_n(1 - x_n).$$

The state is a dimensionless population variable, and μ controls growth and feedback. For small positive x, $x_{n+1} \approx \mu x_n$, while the factor $1 - x_n$ suppresses growth at larger populations. This is an idealized generation-to-generation rule, not a universal population law.

It differs from the continuous logistic differential equation $\dot{x} = rx(1 - x)$. A one-dimensional autonomous continuous flow has strong restrictions absent from a discrete noninvertible map. Therefore their different late-time behaviors are not contradictory.

The graph is a parabola with maximum $\mu/4$ at $x = 1/2$. For $0 \le \mu \le 4$, it maps $[0, 1]$ into $[0, 1]$. In this range a population-fraction interpretation is consistent. Larger or negative parameters can leave that state domain and require a different interpretation.

4 Invariant sets, escape, and numerical safeguards

A set D is forward invariant if $f(D) \subseteq D$. Starting in such a set guarantees that every exact iterate remains there. Proving invariance is a mathematical check distinct from inspecting a few plotted points.

For example, $0 \le x(1 - x) \le 1/4$ on the unit interval proves the logistic range above. In the tent and quartic formulas used in 4A, the bracketed part ranges from zero to one, so $0 \le \mu \le 1$ preserves the unit interval. The parameter normalization differs between maps.

A rapidly escaping trajectory outside the model's intended domain is not by itself bounded chaos. A program may stop when $|x| > 100$ to avoid further growth, but that does not impose a physical boundary, reflect the state, or validate the last computed value. The notebook records the first crossing before it stops. Its

safeguard is designed for the displayed parameter choices, not arbitrary inputs that could overflow inside a function evaluation.

5 Fixed points and local stability

A fixed point satisfies $x_* = f(x_*)$. If $x_n = x_* + \delta_n$ with a small deviation, Taylor expansion gives

$$\delta_{n+1} = f'(x_*)\delta_n + O(\delta_n^2).$$

The multiplier $a = f'(x_*)$ determines local linear behavior:

Multiplier	Local behavior
$0 < a < 1$	Decay without alternating sign
$-1 < a < 0$	Decay with alternating sides
$\lvert a\rvert > 1$	Growth of small generic deviations; repelling fixed point
$\lvert a\rvert = 1$	Linear test is inconclusive; examine higher-order terms

For $a = 0$, higher-order terms control a strongly attracting local convergence. Stability is a response to perturbations: starting *exactly* at a repelling fixed point can still keep the exact trajectory there indefinitely. Local attraction also does not guarantee attraction from every starting state; its basin is the set of states that approach it.

6 Logistic fixed points and the first instability

Solving $x = \mu x(1 - x)$ gives $x_* = 0$ and, for nonzero μ, $x_* = 1 - 1/\mu$. The derivative is $f'(x) = \mu(1 - 2x)$.

At zero the multiplier is μ, so the zero population attracts for $0 \le \mu < 1$ in the physical parameter range. For the nonzero fixed point it is

$$f'(x_*) = 2 - \mu.$$

The nonzero point lies inside $(0, 1)$ and is linearly attracting for $1 < \mu < 3$. At $\mu = 2.8$, $x_* \approx 0.642857$ and multiplier -0.8: convergence alternates around the fixed point. At $\mu = 3.8$ the multiplier is -1.8, so that fixed point is repelling.

The nonzero branch loses linear attraction as its multiplier passes through -1 at $\mu = 3$. This is the first period-doubling threshold. The linear test at the threshold itself is marginal; it should not simply be classified from $|a| < 1$ or $|a| > 1$ alone.

7 Periodic orbits and period doubling

A period-p orbit satisfies $f^{\circ p}(x) = x$ and returns to its starting state after p updates, with no smaller positive return time. A fixed point has period one. A period-two orbit alternates between two distinct values.

Applying the chain rule around a cycle gives its multiplier

$$A_p = \prod_{j=0}^{p-1} f'(x_j).$$

The cycle attracts nearby states if $|A_p| < 1$. A periodic orbit can be stable even after the original fixed point becomes unstable. Counting the intersections of f with the diagonal finds fixed points, not every periodic point; a period-two orbit appears as fixed points of $f^{\circ 2}$.

As the logistic parameter rises, a familiar route involves period two, four, eight, and further doublings. Chaotic parameter ranges also contain periodic windows. Thus increasing μ does not produce a simple monotonic progression from regularity to permanent irregularity.

8 Cobwebs: turn iteration into geometry

8.1 Forward construction

Draw $y = f(x)$ and the diagonal $y = x$ on the same axes. Starting with current horizontal coordinate x_n, move vertically to $(x_n, f(x_n))$. Then move horizontally to the diagonal, giving $(f(x_n), f(x_n))$. Its horizontal coordinate is now the next state x_{n+1}.

Repeat these two segments. The notebook begins its drawing at $(x_0, 0)$ to position the first vertical line; that plotting convention does not make zero the initial state. A contracting staircase suggests an attracting intersection, while a cycle can generate a repeating polygonal path.

8.2 Inverse construction and branch selection

One can instead move vertically to the diagonal and horizontally to the original curve by solving $f(x) = y$. This iterates a selected inverse branch. For $f(x) = x^2$ on $[0, 1]$, that branch is $x = \sqrt{y}$.

Forward squaring makes $0 < x < 1$ decrease toward zero. Inverse iteration by the square root makes it increase toward one. Plotting the inverse staircase against $y = x^2$ does not establish attraction to one under forward squaring. The same square-root sequence can also be drawn as an ordinary forward cobweb against $y = \sqrt{x}$.

Squaring on all real numbers is not one-to-one, and the logistic map also has multiple inverse branches. An inverse construction must specify its domain and branch; the forward-iteration interpretation cannot be assumed unchanged.

9 Other map families and parameter conventions

9.1 Ricker or ecological map

The rule $f(x) = xe^{\mu(1-x)}$ preserves positivity from a positive state. The exponential growth factor can be large at low population and suppress a large population sharply, producing boom-and-crash behavior.

Its positive fixed point is one, with derivative $f'(1) = 1 - \mu$, so it is linearly attracting for $0 < \mu < 2$. These thresholds differ from the logistic map's. A positive state is not necessarily restricted below one: one is a characteristic population scale, not a hard upper bound.

9.2 Gaussian map

The notebook uses $f(x) = e^{-3x^2} + \mu$, where the parameter is an additive shift. Every real input maps into $(\mu, \mu + 1]$. For the large positive shifts in the exercise, the next state is large enough that the exponential becomes tiny. Subsequent values settle close to μ.

Its derivative is $-6xe^{-3x^2}$. The rapid convergence at the displayed offsets is consistent with a small derivative there. Reusing a chaotic logistic-map parameter value does not imply chaos in this differently parameterized Gaussian rule.

9.3 Tent and quartic maps

The notebook's tent rule is

$$f(x) = \mu\,[1 - 2|x - 1/2|],$$

with slopes $+2\mu$ and -2μ on the two branches. Its peak height is μ. Some books parameterize the tent map by its slope instead, so their parameter is 2μ, not μ.

The quartic rule is

$$f(x) = \mu[1 - (2x - 1)^4], \qquad f'(x) = -8\mu(2x - 1)^3.$$

Both preserve $[0, 1]$ for $0 \le \mu \le 1$, but their differing derivatives produce different dynamics. The large $\mu = 2.8, 3.3, 3.5, 3.8$ exercise values violate that domain condition for these formulas; the resulting escape is a

separate question from bounded dynamics.

10 Sensitivity, chaos, and finite precision

10.1 Local stretching along an orbit

For two close trajectories, before their separation becomes large,

$$\delta_n \approx \delta_0 \prod_{j=0}^{n-1} f'(x_j).$$

An average logarithmic stretching rate is

$$\Lambda \approx \frac{1}{n} \sum_{j=0}^{n-1} \ln |f'(x_j)|.$$

With the appropriate long-time limit and trajectory assumptions this gives a Lyapunov exponent per iteration. A positive value indicates sensitive dependence along that orbit. The notebook does not calculate this exponent; it only plots a finite pair separation.

On semilog axes an approximately exponential growth stage may look straight. Once two logistic trajectories are far apart within $[0, 1]$, their separation is bounded by one and need not keep increasing. Transients, finite precision, exceptional orbits, and periodic windows limit what a short plot establishes.

10.2 Exceptional exact starts

At $\mu = 4$, the logistic initial state $x_0 = 0.75$ satisfies $4x_0(1 - x_0) = x_0$. It is an unstable fixed point with derivative -2, but exact initialization does not perturb it. This explains why the revised bifurcation scan uses $x_0 = 0.751$ instead; the other exercises retain their stated initial conditions.

For the tent rule at $\mu = 1$, the chosen $x_0 = 0.75$ follows $0.75 \to 0.5 \to 1 \to 0 \to 0$. This exact exceptional trajectory does not contradict typical chaotic behavior of the full tent map. Its zero fixed point is repelling locally, despite this orbit landing exactly on it.

10.3 What a computer changes

Every stored binary floating-point fraction has finitely many bits. The full tent map shifts and folds those bits, and dyadic initial values can eventually lose all fractional information and reach zero. A deterministic finite-state calculation also cannot produce an infinite sequence of distinct stored states.

These limitations do not make finite-time investigations meaningless. They require distinguishing exact-real mathematical behavior from a particular finite-precision orbit. A diverging or eventually repeating numerical sequence should be interpreted with the rule, domain, and representation in mind.

11 Bifurcation diagrams and late-time sampling

For each parameter, iterate one chosen start, discard a transient, and retain a late segment. Plot each retained value at that parameter. A single stable level suggests period one; two levels suggest period two; branches, bands, and windows reveal changes in the sampled attractor.

The notebook uses 220 parameter values from 2.6 to 4.0. It takes 450 updates for each, discards the first 350 states produced by updates, and retains $x_{351}, \ldots, x_{450}$: 100 plotted states per parameter. Near thresholds, convergence can be slow, so a common fixed discard is an approximation.

Darkness depends on overlap, marker size, parameter spacing, and sample count. It is not a normalized invariant density. A coarse scan can miss narrow windows; one initial state may fail to reveal coexisting attractors in more general maps. A finite diagram is an exploratory view, not a complete theorem about every parameter and start.

12 From theory to a reliable implementation

Before coding, specify the rule, parameter meaning, admissible state domain, and initial state. Decide how many updates to take and which outputs represent transient or late behavior. Keep model cutoffs separate from numerical safeguards.

Check analytic fixed points and multipliers where possible. Test exact simple orbits, including exceptional starts. For logarithmic separation plots, omit exact zero differences rather than silently interpreting them as measured positive numbers. Use longer transients or finer parameter spacing when a result hinges on a small branch or window.

The central distinction is among attraction, periodicity, bounded irregularity, escape, and finite-precision collapse. They are different phenomena and need different explanations. The 4A exercise guide applies these ideas to the notebook's actual plots; the 4B guide develops a different class of stochastic observations.